

Aim: Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

CLI OUTPUT:

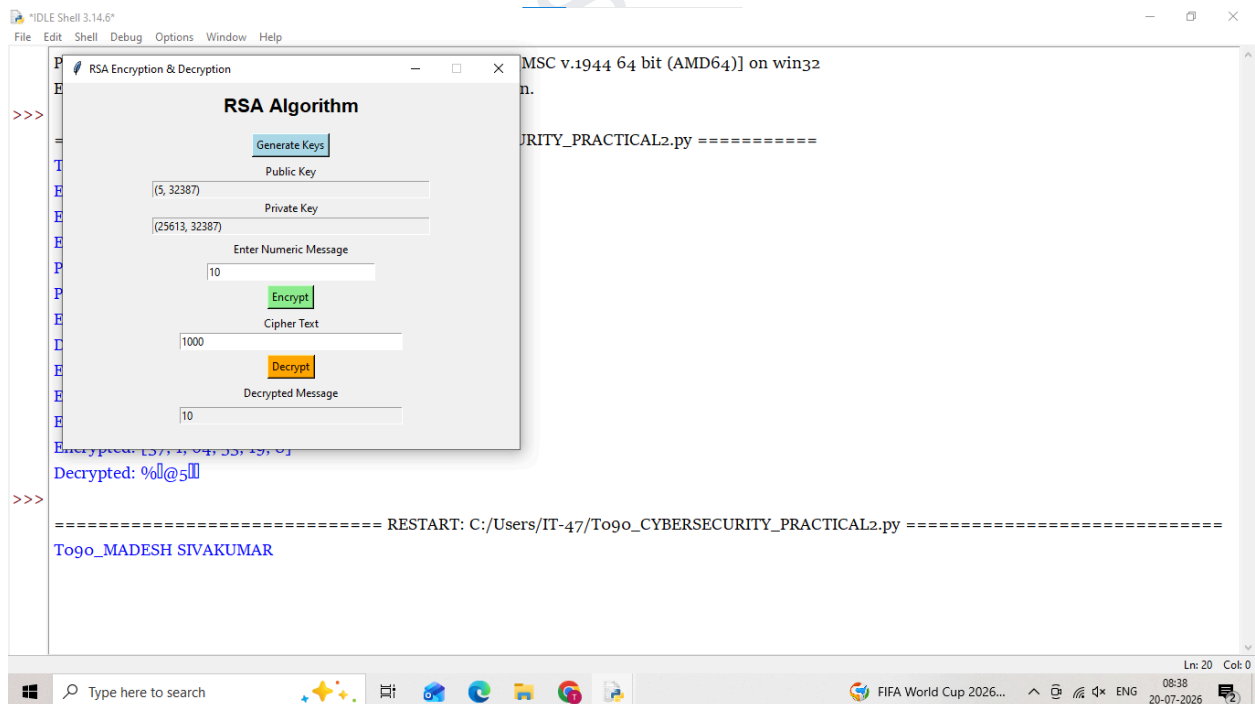


```

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-47/To9o_CYBERSECURITY_PRACTICAL2.py =====
Togo_MADESH SIVAKUMAR
Enter prime p: 69
Enter prime q: 10
Enter message (number): 7
Public Key : (5, 690)
Private Key: (245, 690)
Encrypted : 247
Decrypted : 37
Enter prime p: 8
Enter prime q: 9
Enter name: madesh
Encrypted: [37, 1, 64, 53, 19, 8]
Decrypted: %l@5l
>>>
  
```

GUI OUTPUT:



```

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
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>>>
===== RESTART: C:/Users/IT-47/To9o_CYBERSECURITY_PRACTICAL2.py =====
Togo_MADESH SIVAKUMAR
  
```

The GUI window titled "RSA Algorithm" is overlaid on the shell. It contains the following elements:

- Generate Keys** button: Clicked to generate the public and private keys.
- Public Key** field: Displays (5, 32387).
- Private Key** field: Displays (25613, 32387).
- Enter Numeric Message** field: Contains the value 10.
- Encrypt** button: Clicked to encrypt the message.
- Cipher Text** field: Displays the encrypted value 1000.
- Decrypt** button: Clicked to decrypt the cipher text.
- Decrypted Message** field: Displays the decrypted value 10.